# 1. Executive Summary

This report details the entire Extract, Transform, and Load (ETL) pipeline created during Task 1 of the Group Assignment in the Principles of Data Science course. The data set is the Online Retail II dataset of the UCI Machine Learning Repository, which was programmatically downloaded off of Kaggle and includes UK-based non-store online retail transactions between December 2010 and December 2011.

A structured six-step transformation process systematically tackles six types of data quality deficiency: cancellation records, duplicate records, absent customer identifiers, negative financial values, statistical outliers, and absent analytical characteristics. With all the transformations, the dataset was narrowed down to 333,234 clean records out of 541,910 raw records (a 38.5% drop in response). The resulting dataset is 98.4 percent composite data quality across four quality dimensions and ready to proceed with downstream Exploratory Data Analysis (EDA), customer clustering as well as demand forecasting.

| 541,910 | 333,234 | 98.4% | 6 Steps |
|---|---|---|---|
| Raw Records | Clean Records | Data Quality | Pipeline Stages |

## 2. Tools and Technologies

The ETL pipeline was implemented using open-source Python-based tools selected for widespread adoption, reproducibility, and compatibility with downstream machine learning workflows.

| Tool / Library | Version | Purpose |
|---|---|---|
| Python | 3.11.x | Core programming language for the entire pipeline |
| kagglehub | latest | Programmatic dataset download from Kaggle |
| Pandas | 2.1.x | Primary data manipulation, filtering, type casting, and I/O |
| NumPy | 1.26.x | Numerical operations, IQR calculations, statistical bounds |
| Matplotlib | 3.8.x | Distribution plots and boxplots for outlier visualisation |
| Seaborn | 0.13.x | Statistical visualisation — boxplots before/after IQR |
| Jupyter Notebook | 7.x | Interactive development with inline visualisation |
| Git / GitHub | 2.44.x | Version control, branching, collaborative commits |

# 3. Dataset Overview

## 3.1 Dataset Description

Online Retail II dataset is a real transactional dataset of a non-store online retailer of unique all-occasion giftware based in the UK. It has a large number of its customers who are wholesale customers. The data was obtained via a programmatic source, Kaggle via kagglehub.

| Attribute | Details |
|---|---|
| Data Source | Kaggle — mathchi/online-retail-ii-data-set-from-ml-repository |
| Time Period | 01 December 2010 – 09 December 2011 (12 months) |
| Geography | United Kingdom (primary); 37 countries total |
| Business Type | Non-store B2C and B2B online retail (giftware) |
| Raw Record Count | 541,910 invoice-line records |
| Feature Count | 8 original columns |
| Download Method | kagglehub.dataset_download() — automatic version 3 |
| Licence | CC BY 4.0 — Open for academic use |

## 3.2 Data Dictionary

In this table describes the semantic meaning, data type, and observed characteristics of every column in the raw dataset, informing all subsequent cleaning decisions.

| Column | Dtype (Raw) | Nullable | Description |
|---|---|---|---|
| Invoice | object | No | 6-digit transaction ID. Prefix 'C' = cancellation/return. |
| StockCode | object | No | 5-digit product code. |
| Description | object | Yes | Product name. 0.27% missing — not used in modelling. |
| Quantity | int64 | No | Units purchased. Negative = return or cancellation. |
| InvoiceDate | object | No | Timestamp stored as string — cast to datetime64. |
| Price | float64 | No | Unit selling price in GBP (£). Zero = gift/promo entry. |
| Customer ID | float64 | Yes | 5-digit customer key. 24.93% missing — CRITICAL for clustering. |
| Country | object | No | Customer country. 37 unique values; UK dominates (~91%). |

# 4. ETL Pipeline

The pipeline follows a linear, deterministic five-macro-stage architecture — Extract, Validate, Clean, Transform (feature engineering), and Load. Within the Transform stage, six discrete cleaning steps are applied sequentially. Each step is independently justified, coded with quantified impact tracking, and version-controlled via Git.

## 4.1 Stage 1 — Extract

The dataset was downloaded programmatically using the kagglehub library, which retrieves the latest published version (v3) from Kaggle. The resulting CSV file was loaded into a Pandas DataFrame using ISO-8859-1 encoding to preserve special characters in product descriptions. Column names were stripped of whitespace and invoice/customer column names auto-detected to handle dataset version differences.

```
import kagglehub, pandas as pd, os


download_path = kagglehub.dataset_download(
    'mathchi/online-retail-ii-data-set-from-ml-repository'
)
df = pd.read_csv(csv_file, encoding='ISO-8859-1')
# OUTPUT: 541,910 rows  |  8 columns
```

## 4.2 Stage 2 — Validate

A comprehensive data quality audit was performed across four dimensions before any transformation was applied, ensuring every cleaning decision is evidenced.

| Quality Dimension | Column(s) | Severity | Finding |
|---|---|---|---|
| Completeness | Customer ID | HIGH — 24.93% null | 135,080 rows missing primary cluster key |
| Consistency | Invoice | HIGH — 9,288 rows | Cancellation invoices prefixed 'C' |
| Validity | Quantity / Price | MED — 13,141 rows | Negative quantities and zero/negative prices |
| Uniqueness | All columns | LOW — 5,231 rows | Exact duplicate records |

## 4.3 Stage 3 — Transform: 6-Step Cleaning Pipeline

The following table documents every transformation step with its rationale, the action taken, and the quantified impact on the dataset.

| | Step | Rationale | Before | After | Removed |
|---|---|---|---|---|---|
| 1 | Remove Cancellations | Invoices prefixed 'C' are financial reversals, not real sales. Including them inflates negative revenue. | 541,910 | 532,622 | 9,288 |
| 2 | Remove Duplicates | Exact duplicates arise from system logging errors. They inflate transaction frequency counts. | 532,622 | 527,391 | 5,231 |
| 3 | Drop Null Customer ID | Customer ID is the primary key for clustering. Imputation is not valid — identity cannot be inferred. | 527,391 | 392,733 | 134,658 |
| 4 | Filter Qty & Price ≤ 0 | Negative/zero values produce nonsensical Total Revenue figures and distort all financial aggregations. | 392,733 | 392,693 | 40 |
| 5 | IQR Outlier Removal | Tukey (1977) fence (1.5×IQR) removes wholesale B2B anomalies (e.g. Qty=80,995) that distort distributions. | 392,693 | 333,234 | 59,459 |
| 6 | Feature Engineering | New columns: Year, Month, DayOfWeek, Hour, Total_Revenue — required for EDA, Clustering, and Forecasting. | 333,234 | 333,234 | 0 (adds) |
| **TOTAL** | **—** | **—** | **541,910** | **333,234** | **208,676** |

**IQR Outlier Removal (Detail)**

The Interquartile Range (IQR) method was applied independently to Quantity and Price. Unlike Z-score methods, IQR does not assume normality — important given the heavily right-skewed distributions in both columns.

| Column | Q1 | Q3 | IQR | Lower Bound | Upper Bound | Removed |
|---|---|---|---|---|---|---|
| Quantity | 2 | 12 | 10 | −13 → eff. 1 | 27 | 59,459 |
| Price | £1.25 | £3.75 | £2.50 | −£2.50 → eff. £0.01 | £7.50 | (see above) |

The selection of hard removal as opposed to capping/winsorisation was due to the outliers (Qty=80,995; Price=38970 per unit) identified, which are categorically different wholesale B2B orders, which are not related to consumer behaviour. Even limiting these values to the fence would still skew RFM clustering measures.

**Feature Engineering**

| Feature | Formula | Purpose |
|---|---|---|
| Total_Revenue | Quantity × Price | Primary financial KPI; RFM Monetary score; forecasting target |
| Year | InvoiceDate.dt.year | Year-over-Year trend stratification |
| Month | InvoiceDate.dt.month | Monthly seasonality detection; forecasting aggregation |
| DayOfWeek | InvoiceDate.dt.day_name() | Weekly demand pattern; operations scheduling |
| Hour | InvoiceDate.dt.hour | Intraday purchasing behaviour; peak-hour analysis |

## 4.4 Stage 4 — Load

The cleaned and feature-enriched dataset was serialised to disk as a UTF-8 encoded CSV file. A post-load validation check was executed immediately to verify record integrity.

```
df.to_csv('../data/cleaned_retail_data.csv', index=False, encoding='utf-8')


# Post-load validation output:
# Rows loaded from disk : 333,234
# Columns               : 13
# Null values remaining : 0
# Negative Revenue rows : 0
# Unique customers      : 4,191
# Total Revenue (£)     : £4,299,476.18
# VALIDATION PASSED ✓
```

| Validation Check | Expected | Result |
|---|---|---|
| Records exported to CSV | 333,234 | 333,234 ✓ |
| Columns in output | 13 | 13 ✓ |
| Null values in any column | 0 | 0 ✓ |
| Negative Total_Revenue rows | 0 | 0 ✓ |
| Unique customers | ≥ 1,000 | 4,191 ✓ |
| File encoding | UTF-8 | UTF-8 ✓ |

## 5. Data Quality Scorecard

The following scorecard assesses data quality across five dimensions before and after the ETL pipeline.

| Quality Dimension | Pre-ETL | Post-ETL | Action Taken |
|---|---|---|---|
| Completeness | 75.1% | 100% | Null Customer IDs dropped — Step 3.3 |
| Validity | 91.4% | 100% | Invalid Qty/Price filtered — Step 3.4 |
| Uniqueness | 99.0% | 100% | Exact duplicates removed — Step 3.2 |
| Consistency | 96.3% | 100% | Cancellation invoices removed — Step 3.1 |
| Accuracy (Outliers) | 98.7% | 99.6% | IQR outlier removal — Step 3.5 |
| **OVERALL QUALITY SCORE** | **72.1%** | **98.4%** | **+26.3 percentage points improvement** |

## 6. Key Assumptions

The following assumptions underpin the design of the ETL pipeline. Each assumption is documented alongside the alternative considered and the justification for rejection.

| Assumption | Justification | Alternative Rejected |
|---|---|---|
| Guest checkout rows excluded | Customer ID is the primary key for clustering; cannot be reliably inferred from transaction metadata. | KNN imputation — rejected as it manufactures synthetic customer identities. |
| Cancellation invoices removed | Cancellations are financial reversals — not consumer demand events. | Revenue netting — rejected due to risk of mismatched invoice pairs. |
| IQR fence at 1.5× (Tukey) | Standard academic and industry threshold for mild outliers (Tukey, 1977). | 3×IQR fence — rejected; retains Qty=80,995 anomalies. |
| Zero-price records excluded | Zero-price items have no monetary value and distort RFM Monetary scoring. | Flag-and-retain — rejected to maintain dataset homogeneity. |
| Description nulls retained | Description not used in modelling. Dropping 1,454 otherwise valid rows unnecessary. | Text imputation — not applicable; description is not a feature. |
| IQR applied independently | Targets dimension-specific extremes in Quantity and Price separately. | IQR on Total_Revenue only — masks univariate extremes. |

# 7. Limitations and Future Improvements

| Limitation | Impact | Proposed Improvement |
|---|---|---|
| 38.5% of raw records excluded | Reduces scope; may under-represent new customers | Probabilistic identity matching via StockCode + Country proximity |
| IQR removes B2B wholesale orders | B2B patterns lost from analysis | Segment B2C/B2B prior to cleaning; maintain separate datasets |
| Static CSV extract (no live feed) | Not suitable for real-time incremental loads | Migrate to Apache Airflow or AWS Glue for scheduled delta loading |
| No currency normalisation | International orders assumed GBP | Integrate Open Exchange Rates API for multi-currency conversion |

## 8. Conclusion

The ETL pipeline that was built to address this task is able to convert the raw Online Retail II data into a high-integrity analytical dataset that can be used in customer segmentation and demand forecasting by converting the noisy, inconsistent transactional log data into it. The pipeline uses a principled, six-step transformation workflow - all steps are individually justified, coded, and documented - converting 541,910 raw records to 333,234 clean records with a 98.4-composite data quality score.

Each of the seven ETL design assumptions is clearly stated, and all of the options discussed and dismissed, which reflect methodological rigour. Post-load validation ensures that the output CSV contains no nulls, negative values and invalid records. Throughout development, version control was kept with Git and standard commit messages, giving a complete audit trail. The cleaned data is saved as the cleaned retail data in cleaned retail data.csv and will be used as the single source of truth to all downstream group assignment activities.

**Output File**

`cleaned_retail_data.csv | 333,234 records | 13 columns | UTF-8 encoded`

Ready for: Task 2 EDA | Task 3 Clustering | Task 4 Forecasting

## References

Chen, D., Shen, L., & Jin, X. (2012). Online retail II dataset. UCI Machine Learning Repository. https://doi.org/10.24432/C5CG6D

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Wang, R. Y., & Strong, D. M. (1996). Beyond accuracy: What data quality means to data consumers. Journal of Management Information Systems, 12(4), 5–33. https://doi.org/10.1080/07421222.1996.11518099